IONIC AND COVALENT COMPOUNDS 12

Only the noble gases exist as discrete (separate) atoms. Imagine that a complete outer shell of electrons forms a kind of 'case' around the atom. Then think of that case as complete and enclosing the whole unit. The atoms of all other substances that are not noble gases become bonded in some way as a result of having incomplete outer 'shells'. These incomplete shells need to gain, lose or share a specific number of electrons to form a full outer shell. When the shell is full, the substance reaches the stable electronic configuration of the noble gases. The type of bonding present in a compound will affect its properties (the way it looks, feels and behaves). The main types of bonding are ionic, covalent and metallic.

Ionic compounds

These compounds are also known as **electrovalent compounds**. Ionic compounds are composed of ions (charged particles) and electrically-neutral molecules. Ions are formed by electron transfer between metals and non-metals. The ions that are formed are attracted by the opposite electric charges. These ions quickly arrange themselves into giant structures in which each positive ion is surrounded by negative ions and vice versa. Once ions have been formed, they do not have the same properties as their parent atoms. This is because they now have a different electronic structure to the parent atom and therefore different chemical properties. Ionic compounds are generally solids (often crystalline).

1 Circle the ionic compounds in the list below.

NaF, H_2O, $MgCl_2$, H_2, CaI_2, PCl_3, LiBr, NH_3

In ionic compounds, the positive ions (**cations**) have been formed from metals and the negative ions (**anions**) from non-metals. For example, sodium chloride, NaCl, is made up of sodium (Na^+) and chloride (Cl^-) ions. Its structure is a repeating three-dimensional lattice that is held together very tightly because the positively-charged sodium ions are attracted to the negatively-charged chloride ions.

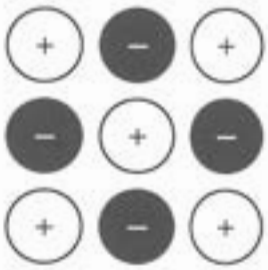


Figure 1

Each ion is surrounded (in three dimensions) by six oppositely-charged ions. The shape of the lattice is a cube. The attractive force between ions is large (and non-directional) and a lot of energy is needed to move the ions apart. This tightly-packed structure makes the bonds harder to break, resulting in high melting and boiling points. Figure 1 shows part of one layer of an ionic lattice.

2 Redraw figure 1 to show how sodium and chloride ions might be arranged in a layer of an ionic lattice.

3 Add to your diagram to show how the next layer of ions might be arranged.

The formula for an ionic compound is an **empirical formula**; it shows the ratio of ions present, not the actual numbers.

4 Why is the empirical formula given for an ionic compound and not the actual numbers of ions present?

5 Give the electrical charge on an anion.

6 Give the electrical charge on a cation.

When looking at the solubility of substances, the phrase 'like dissolves like' is helpful. Many ionic compounds are soluble in water, which is a polar solvent. Polar substances have electrically-charged ends. Figure 2 shows a simple polar substance. One end is slightly positive (shown as δ+) and the other end is slightly negative (shown as δ–). The symbol δ (delta) stands for a small change. Ionic compounds are usually insoluble in organic (carbon-based) solvents, because these solvents are covalent substances, not polar substances.



polar substance

Figure 2

An electric current is a flow of electrons or ions. The outer electrons in ionic compounds are used to give the ions their noble gas type structures. These electrons cannot move away from their ions. Although ionic compounds are composed of charged particles, they do not conduct electricity when they are solid. When molten (fused) or in solution, the ions are no longer held together as strongly and they are free to move. If electrodes are placed in such a liquid, the ions allow a current to flow and this is accompanied by decomposition (breakdown) of the substance. This is called **electrolysis**.

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7 What does it mean to say an ion in a compound has reached a noble gas type structure?

8 Name two types of bonding apart from ionic.

9 Name the only types of elements that exist as separate (non-bonded) atoms.

10 Why do ionic compounds tend to have high melting points?

Figure 3 shows how the ions move apart when an ionic compound is dissolved. (Note that the water molecules are not shown in the diagram.) If positive and negative electrodes are placed in an ionic solution, the ions will be affected.

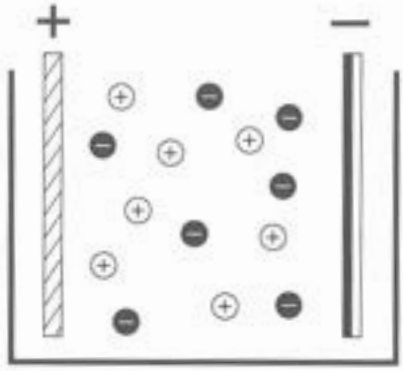


Figure 3

11 Using figure 3 as a model, draw a diagram to show how ions would move in a solution if electrodes were present.

12 Why can ionic compounds only conduct electricity when they are in liquid form?

13 Match the following terms with the definitions below:
cation, electrolysis, empirical, ionic, octet, polar, solvent, valence

a The type of bond formed from the electrostatic attraction between oppositely-charged ions: _____

b An ion that would be attracted to the cathode in electrolysis: _____

c Outermost electrons that are transferred or shared in bonding: _____

d The process that uses electricity to split up an ionic compound when it is in liquid form: _____

e A formula that shows the ratio of elements present: _____

f A liquid that can dissolve other substances: _____

g The full outer shell of electrons that is present in the noble gases from neon to radon: _____

h A substance that is not ionic but has ends that have a slight difference in electrical charge: _____

Covalent substances

Compounds that contain only non-metals usually consist of molecules. They can be molecules of single elements (e.g. oxygen, chlorine, carbon) or compounds of different elements. Covalent substances can be divided into two groups: small (discrete) molecules and giant molecules (macromolecules/covalent lattices). In both cases, the atoms within a molecule are bonded to each other by strong covalent bonds. Covalent bonds contain shared electrons. Small molecular substances often have between two and twenty atoms per molecule. Giant molecules have many millions of atoms per molecule. Remember, in ionic compounds, one atom tends to give up 'spare' electrons to another atom that only needs a few electrons to fill its outer shell. In covalent bonding, one atom does not 'take' electrons from another, they are shared.

Small molecules are discrete; they tend to be separate and can move independently of each other. They have low fixed points (melting points and boiling points). This is because there are only relatively small attractive forces between electrically neutral molecules. Tetrachloromethane is a typical molecular covalent substance. It is a liquid with a low boiling point. The structure of tetrachloromethane is shown in figure 4.



Figure 4

14 What is the formula of tetrachloromethane?

15 Review your notes on the previous page explaining how ionic compounds are formed, then look at the tetrachloromethane molecule in figure 4.

a Are there any metals present in this molecule?

b What does the composition of the tetrachloromethane molecule suggest to you about the type of bonding occuring between the atoms (ionic or covalent)? Explain your answer.

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Substances made up of molecular covalent molecules often have a smell; ionic and macromolecular substances generally do not. Visualise a smell as molecules that get up your nose; from this it is clear that molecular covalent substances can put molecules into the vapour stage more easily than other substances can. (It is breathing in vapours that allows us to smell things.)

16 Why do molecular covalent substances have low boiling points?

Diamond is made up of carbon atoms arranged in a lattice. Figure 5 shows the structure of diamond. Each carbon atom is attached to another four carbon atoms. The number of the bonds and the strength of the bonds make diamond a very hard substance with a high melting point.



Figure 5

17 Discuss the statement: *A perfect diamond is one single, giant molecule.*

It is much easier to melt a substance that has a small molecular structure than a substance that has a covalent lattice. There is no need to break the strong covalent bonds inside the molecules; rather, separate molecules that are only weakly attracted to each other. It is important to distinguish between **intermolecular** forces and **intramolecular** forces.

Intramolecular forces are those inside the molecule and they are strong covalent bonds (*intra* means within, inside). Intermolecular bonds are between individual molecules and they are weak (*inter* means between). Figure 6 shows the bonding in methane (natural gas), a molecular covalent substance.

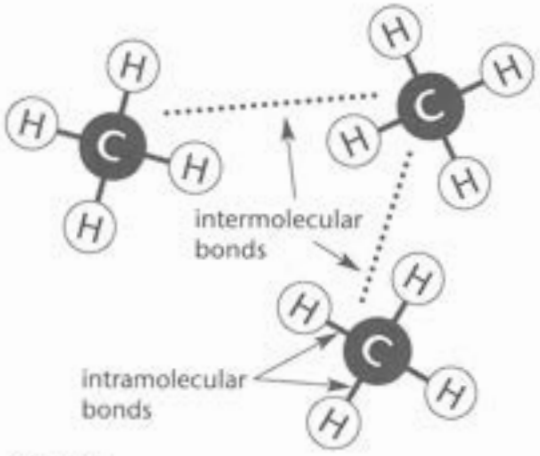


Figure 6

18 Give the formula for methane.

19 Melting and boiling points are sometimes called 'fixed points', because they occur at fixed temperatures. Will methane have low or high fixed points? Explain your answer.

Covalent compounds do not conduct electricity in solid, liquid or gaseous states. There are no ions or free electrons to carry a current. Many covalent substances are insoluble in water (a polar substance), but soluble in other covalent liquids such as ethanol (C_2H_5OH) and tetrachloromethane (CCl_4).

20 What type of bonding is present in the ingredients of solid air-fresheners? Explain your answer.

21 Give the names and formulae of two covalent liquids.

22 Explain why covalent substances do not conduct electricity.

23 Suggest why many covalent substances are soluble in other covalent liquids.

24 In chemistry, what does it mean when a compound is said to be 'organic?

Figure 7 shows two ammonia molecules. Ammonia is a covalent compound. It is a gas at low temperatures but is easily liquefied.

25 Give the formula for ammonia.

26 Copy figure 7. On your diagram show the intermolecular and intramolecular bonding. (Hint: see figure 6.)

27 Ammonia is very soluble in water. Does this mean that ammonia is polar or non-polar? Explain your answer.

28 Make a list of the similiarities and differences between ionic and covalent bonding.

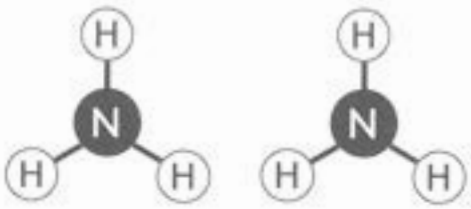


Figure 7

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